

The Fatal Hydrophobic Collapse: Sickle Cell Anemia

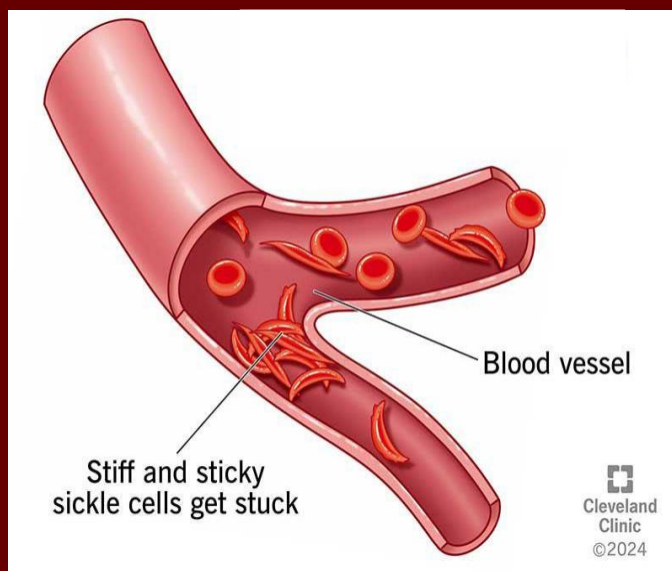
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INTRODUCTION

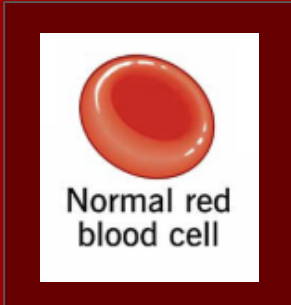
- Sickle Cell Anemia (also known as Sickle Cell Disease) is a genetic disorder coming from a mutation in hemoglobin. The amino acid changes from Glutamic Acid, which is a hydrophilic acid, to Valine, a hydrophobic one.
- The change causes the affected cell to have an abnormal hydrophobic collapse as the Valine hides from the water and blood cell collapse.
- The rigid and sticky cells to stick to the blood vessels, blocking red blood cells from getting through the vessel.
- This causes a range of symptoms like severe pain episodes (crises), chronic fatigue, jaundice, paleness, shortness of breath, etc.

Fig 1: Sickle Cells block the red blood cells



PURPOSE

To compare and contrast the Wild Type (WT) and Mutant (M) versions' 'compactness' through the Radius of Gyration, SASA, DSSP and the RMSD.



METHODOLOGY

1. Generating the topology file:

- Find the desired protein
- Upload it into the google colab
- Convert it into a clean pdb file, then execute pdb2gmx
- Choose a force field (I chose #8:CHARMM36)

1. Defining the box and solvate:

- Define the box using [!gmx editconf] command (should generate a box.gro file)
- Fill the box with solvent using [!gmx solvate] command (generates a solvent.gro file)
- When using VMD, you can input 'pbc box' into the code to generate the box

1. Adding Ions:

- There should be a net charge of 8 on the topology (topol.top) file
- Assemble the .tpr file
- Then you would use genion for the ions.tpr file

1. Energy Minimization:

- Assemble the binary input using [!gmx grompp] (using minim.idp file)
- Then carry out the EM in [!gmx mdrun] (you should get 4-5 files; em.tpr, em.log, em.edr, em.trr and em.gro)
- Then you can do a bit of analysis with the em.edr file for potential using [!gmx energy]

1. NVT + NPT Equilibration:

- Use the [!gmx grompp] and [!gmx mdrun] commands like the EM
- Analyze the temperature for both mutant and wild types [!gmx energy] and use plot.svg to plot them both (these aren't shown)
- Do the same thing for the NPT equilibration, with pressure and density,

1. Production Run:

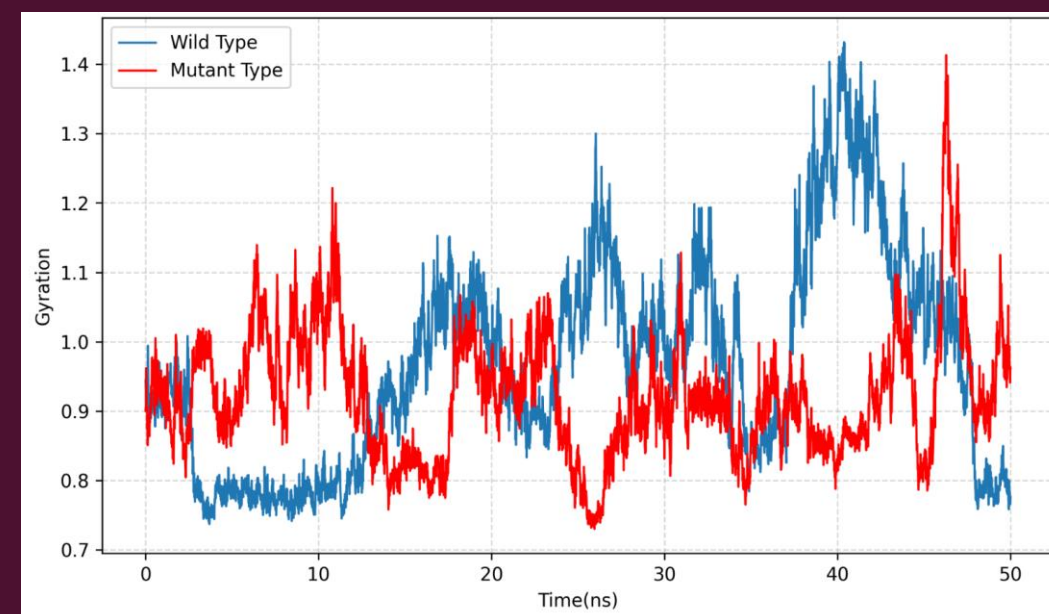
- Use [!gmx grompp] command including the md.mdp file, npt.gro, topol.top file
- Run [!gmx mdrun] (This should take 1 to 1.5 hours on GPU, on CPU it takes 5-6 hours)

1. Data Analysis:

- RMSD: Use the md.tpr and md.xtc files, choose 4 twice, then run [!gmx rms] command and plot it.
- Rg: Run [!gmx gyrate] using the same .tpr and .xtc files
- SASA: Use [!gmx sasa] command, then plot using xvg file
- DSSP: Use [!gmx dssp] command, then plot using plot.svg
- Hydrogen Bonds: Using the [!gmx hbond] commands, do the MainChain+H group, then SideChain group.

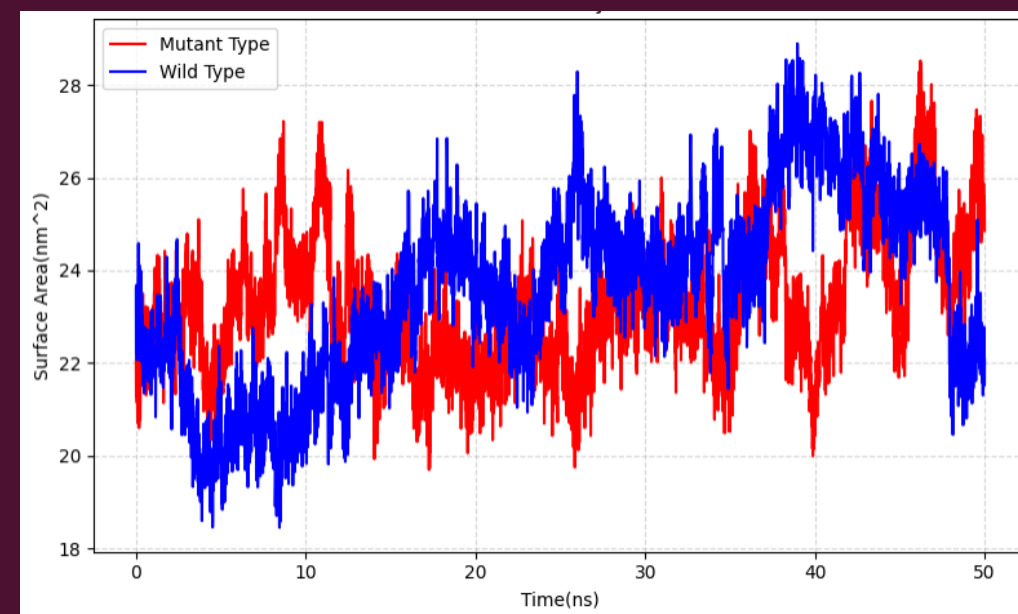
RESULTS

Figure 2: Radius of Gyration Analysis



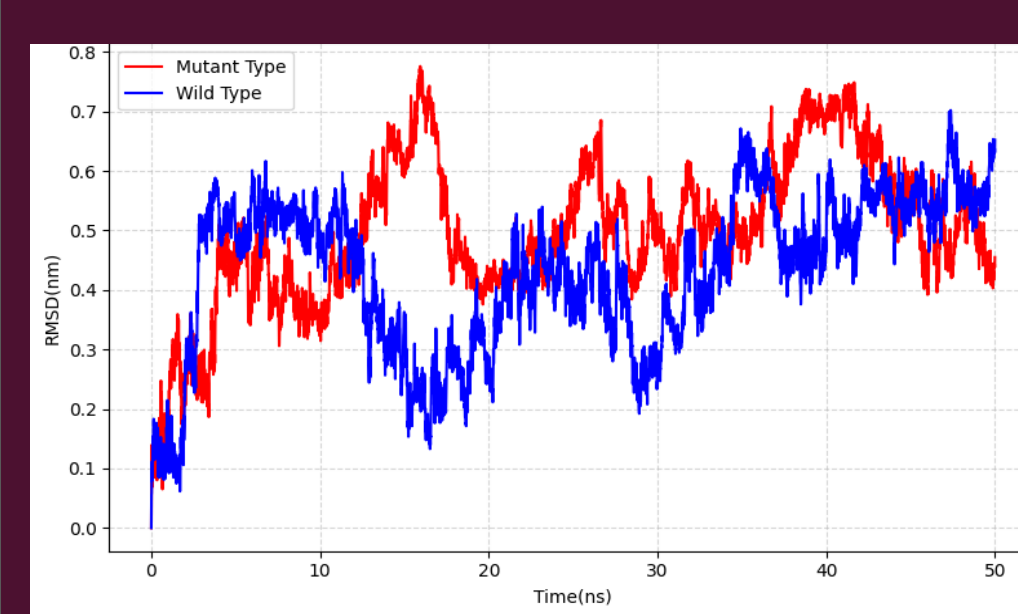
- Initially, the Wild type is fairly stable and compact at 0.9 nm, then dips to about 0.75 nm and settles to be 0.8-1.1 nm. At around 39-45 ns, there is a big expansion (meaning it's swelling/expanding) to around 1.45 nm.
- The Mutant type also starts at around 0.9 nm, but fluctuates significantly during the entirety of the experiment between 0.8-1.2. It also has a spike near 1.4 nm, but its later than the Wild type at 45 ns.

Figure 3: SASA Analysis



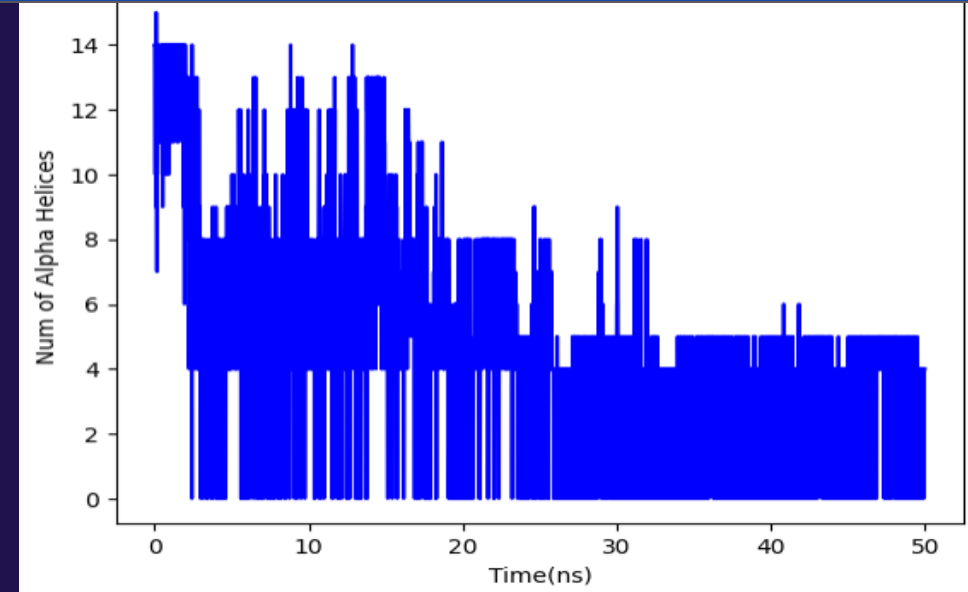
- They both start at around 20-24 nm^2
- The Wild Type is stable and has a higher SASA, and the Mutant Type has lower SASA and is unstable

Figure 4: RMSD Analysis



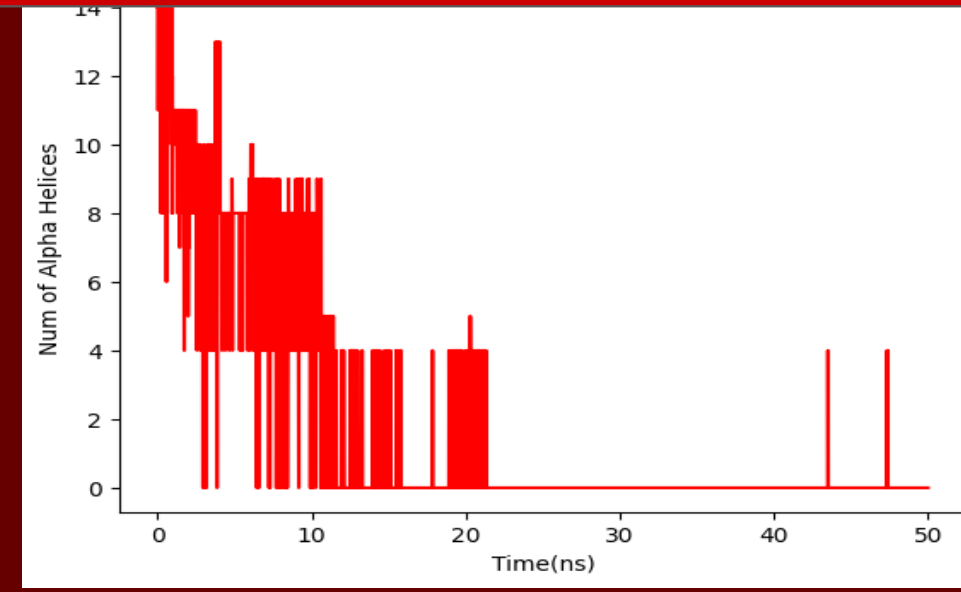
- The Wild Type starts at ~0.1 nm and rises quickly to ~0.4 nm. Then it peaks at ~15 ns with 0.75 nm and ~38-41 ns with around 0.7 nm. It settles back into fluctuations of 0.4-0.6 nm.
- Eerily enough, the Mutant Type is similar to the Wild Type, but are less sudden. This can suggest that it's not affecting the global RMSD as much.

Figure 4 : DSSP Wild Type Analysis



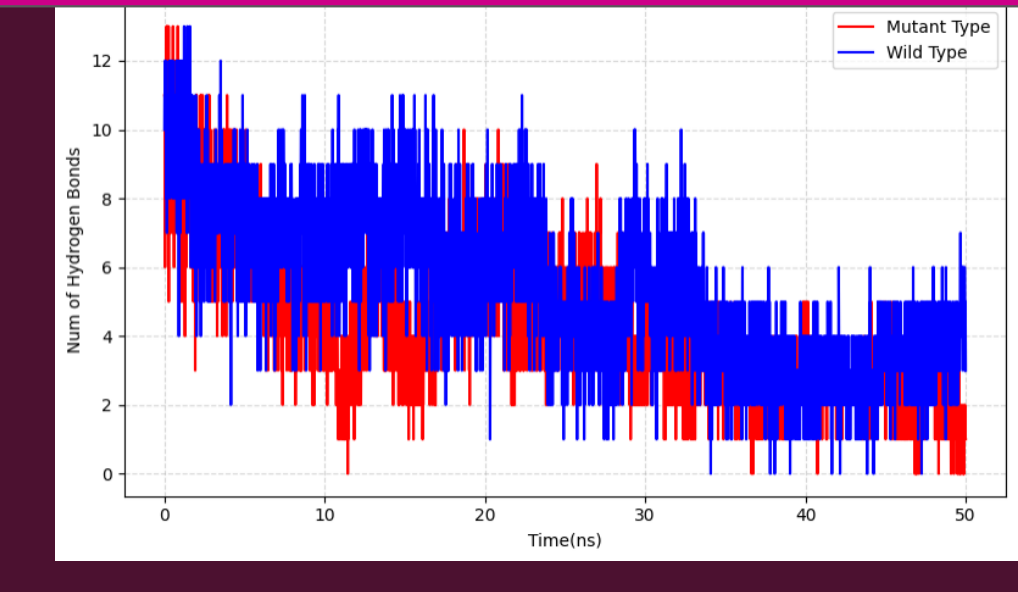
- Starts off with 13-15 Helices, slow and gradual speed of loss
- There are 4-5 Helices that are retained by the simulation's end
- Relatively Stable throughout the simulation

Figure 5 : DSSP Mutant Type Analysis



- Starts off with 13-14 Helices
- Dramatic speed of loss after 20 ns, with brief spikes between 40-50 ns.
- There are no helices that are retained
- Instability throughout the simulation, with the unfolding happening after 20 ns

Figure 6 : Hydrogen Bond Analysis

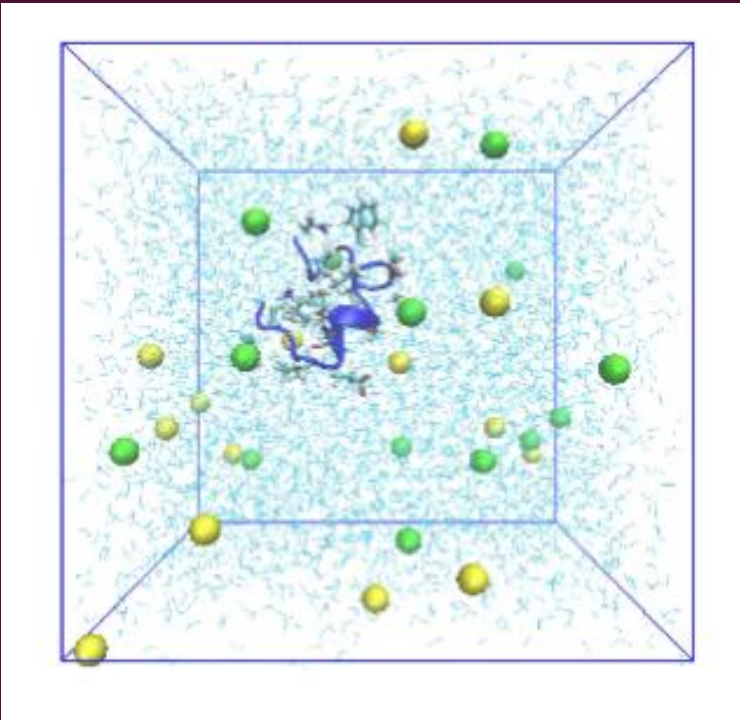
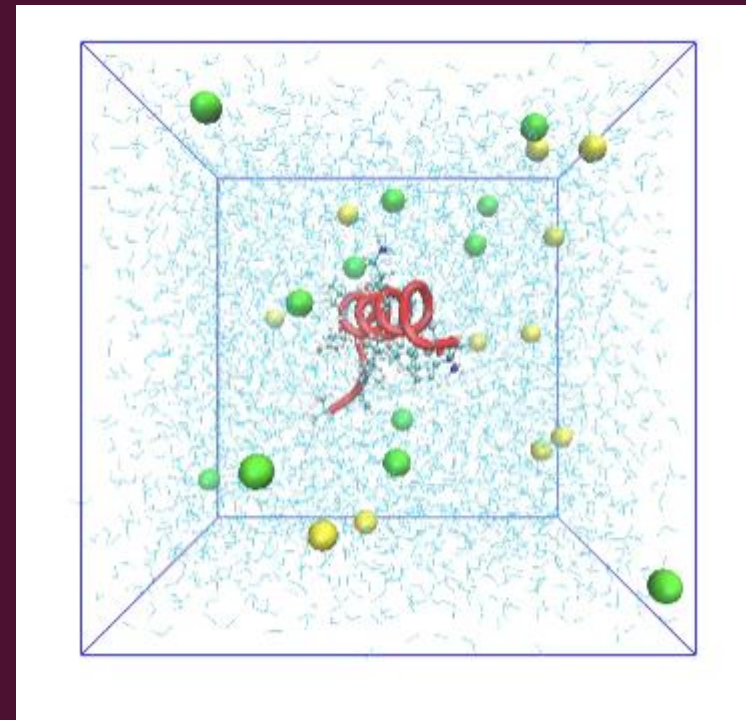


- Both start with ~12-13 hydrogen bonds and slowly lose them over time.
- The Wild type maintained around 3-8 hydrogen bonds, but the Mutant type has 0-2 hydrogen bonds at the end of the simulation, making it unstable.

Figures 7&8:

These are the VMD Visuals (Mutant, then Wild):

Legend:
- Protein = Red
- Sidechain = Blue
- Water = Cyan
- Sodium Ion = Yellow
- Chlorine Ion = Green



TERMINOLOGY

Rg (Radius of Gyration): A number telling you how compact or spread out a protein is

SASA (Solvent Accessible Surface Area): Measures how much protein is touching the solvent

DSSP: A computer Program reading the Secondary structures of a protein and labels them

RMSD (Root Mean Square Deviation): A number telling you how different protein shapes are from one another

ACKNOWLEDGEMENTS

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WEBSITES

For this experiment, the software I used were the following:

Protein Database
AlphaFold
Google Colab
GROMACS
VMD

LIMITATIONS

- Every time we needed to reload the Colab, we had to install and Verify GROMACS and Mount the Google Drive into the Colab. Every. Time.
- Installing the ions needed for both the wild and mutant type took a lot of time.
- When analyzing some of the graphs for the RMSD or SASA for example, the production.xtc file I used were too small to be used and didn't work. I fixed it using the production.gro file containing everything that i did up to that point.

CONCLUSION

In the end, there are key differences between the regular Wild Type Hemoglobin and the Mutated Sickle Cell type, and the data shows that clearly:

- Rg**: The Mutant Type was less stable and more jittery than the Wild Type, and had a harder time holding its shape. This is consistent with the disease; when the amino acid is changed, the E6V causes the cell to collapse and clumping because of E6V's hydrophobic properties.
- SASA**: Both of them were fairly noisy over time, but the difference lies in how it's presented: the Wild type was more exposed to the solvent and trends higher later on with several spikes near the end. However, the Mutant type has a lower SASA, meaning it wasn't as exposed to the solvent, and it was collapsing into a compact state and occasionally flickering open.
- Hydrogen Bonds**: Both start at 12-13 hydrogen bonds and slowly decrease over time. However, by the end of the simulation, it shows the Wild type having more hydrogen bonds than the mutant type.
- DSSP**: Here's the main difference: the Wild type starts at around 13-15 helices that steadily decreases over time, but the Mutant type dramatically decreased to 0 helices after 20 ns and unfolded completely, heavily supporting the SASA, Hydrogen Bond and Rg data.
- RMSD**: For this piece of data, both of them didn't have as much of an effect on the global RMSD for both the Wild and Mutant type. Both of them start and finish in roughly the same places at the same time, but the Mutant type was a little later.

FURTHER RESEARCH IDEAS

- Hydrophobic SASA Analysis of the Sickle Cell**: The SASA chart in this simulation was for the Total SASA, but the Hydrophobic SASA can test whether the valine patch drives the aggregation.
- Comparing other Sickle-cell Mutations**: There are other mutations rather than just E6V, and it would be interesting to compare between different varieties of the same disease.
- Free Energy Calculations on Sickle-cell Mutations**: Using PBSA/MM or Umbrella Sampling, we can potentially quantify *how much* the mutation destabilizes the protein.
- Testing Therapeutic hypotheses**: Using a small molecule or known drug, we could figure out if this can truly rescue the mutant's stability (and we can do different varieties of the same mutation)